LEARN GIT FUNDAMENTALS

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Project Overview: Learning Git Version Control

Goals and objectives

In this project, I'm building a git repo so that I can leran the git essentials of creating a repo, a branch and handle merge conflicts.

Setting Up the Development Environment

```
PS C:\WINDOWS\system32> git --version
git version 2.53.0.windows.1
PS C:\WINDOWS\system32>
```

Configuring Git and Creating the Learning Log

Setting up Git identity and project file

In this step, i'm setting up my git identity so that i can update files and Git will know that i made the changes.

```
git-learning-log ×
C: > Users > MSI > git-learning-log > git-learning-log > # My Learning Log > ## Goals
# My Learning Log

## About This Project
I'm learning Git and version control to track my work.

## Goals
- Understand how commits work
- Learn branching and merging
- Push my work to GitHub
```

Understanding global Git configuration

The --global flag means my user.nzmr znd my user.email will be used globally across repos. I used a specific email because that's the email address that is associated to my github account.

Initializing a Repository and Making Commits

Creating the first commits

In this step, I'm setting up a git repository so that I can make changes and commit it using git tools.

```
C:\Users\MSI\git-learning-log>git log --oneline
edd530c (HEAD -> master) Add What I Learned section
908a38a Add initial learning log
```

Understanding git add vs git commit

The difference between git add and git commit is that git add does include the file to the stagine while git commit commits the changes along with a message.

Branching and Working in Parallel

Creating and switching branches

In this step, I'm creating a new branch so that I can make a commit on the new branch. This will so help me understand what will happen when I make a conflicting commit on main that I'll have to handle.

Merging Branches and Resolving Conflicts

Merging feature branches into main

In this step, I'm merging two branches with comfmicting changes so that I can read and understand Git's conflict markes and resolve and complete the merge commit.

```
C:\Users\MSI\git-learning-log>git log --oneline --graph
*   6fc87dc (HEAD -> master) Merge add-resources branch and resolve conflict
|\
| * 0cdcc42 (add-resources) Expand learning notes on branches
* | f441b24 Rewrite learning notes with time machine analogy
|/
* edd530c Add What I Learned section
* 908a38a Add initial learning log
```

Interpreting Git conflict markers

The conflict markers show me what the current changes are, what the incoming changes are, and it allows me to modify it as needed for my fi al file.

Publishing to GitHub

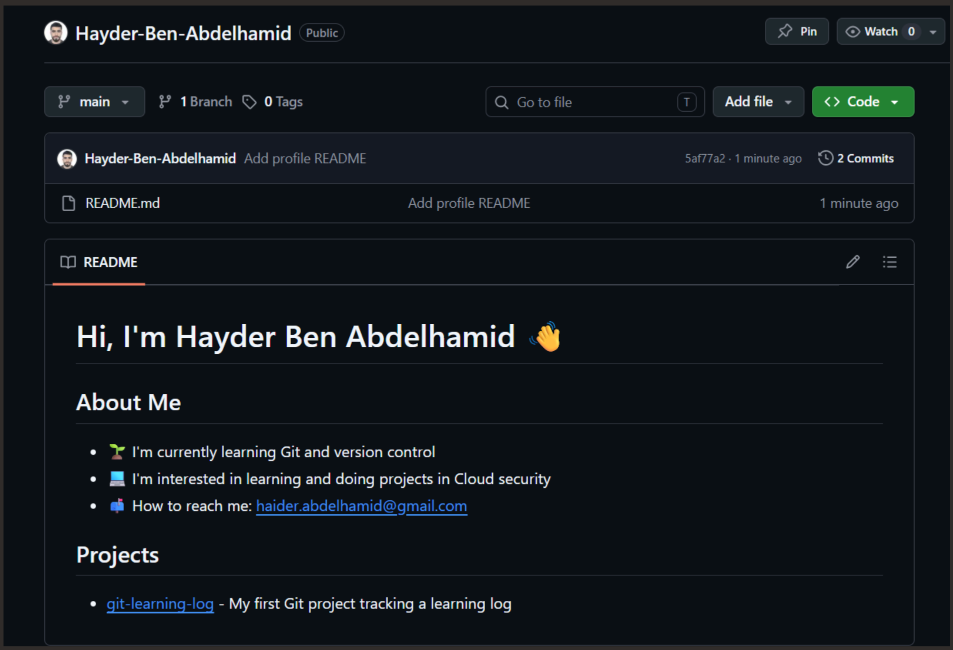
Creating a remote repository

In this step, I'm setting up a public GitHub repository so that I can push my local repository into GitHub and verify my commit history and branches are visible online.

Pushing local commits to GitHub

I ran git push to upload my local commits and branch history from my computer to GitHub so that they are backed up safely online.
I needed a remote because Git needs a destination address to know exactly where to send my files when i run the push command.

Bonus: Building a GitHub Profile README



Commands used to publish the Profile README

In this project extension, I used the commands add, commit and push to push my README to the public guthub profile.